

Exam Date & Time: 30-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
 BTech V Semester - Make up Examination - December -2024
Number Theory and Cryptography [IT-3151]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Solve the following
 - a. $\Phi(2^3 \cdot 5^2 \cdot 7^5 \cdot 11^3)$
 - b. LCM (60, 72, 108)
 - c. Caley Table Z_7^*

A) (5)

B) Analyse the uniqueness of the prime factorization theorem. That is, if n has two different prime factorizations, show that they must be the same up to the order of the factors. (5)
- 2) Show that the inverse of 5 modulo 101 is 5^{99} (5)

A)

B) For each of the following equations, find an integer x that satisfies the equation.

 - a. $4x \equiv 2 \pmod{3}$
 - b. $7x \equiv 4 \pmod{9}$
 - c. $5x \equiv 3 \pmod{11}$

(5)
- 3) Prove that triplet $\{Z_n, +_n, *_n\}$ for non-prime n is a commutative ring (5)

A)

B) Show that an integer N is congruent modulo 9 to the sum of its decimal digits. For example, $723 = 7 + 2 + 3$ $K=12 = 1 + 2 = 3 \pmod{9}$. This is the basis for the familiar procedure of "casting out 9's" when checking computations in arithmetic. (5)
- 4) You are given the following ciphertext: VGRJYQWZ and the key matrix used for encryption is

A)

	6	24	1
$K =$	13	16	10
	20	17	15

Further the key is modified in such a way that, (5)

Row 1 = Row 1 * (multiplicative inverse of 565 mod 79)

Row 2 = Row 2 * $\text{pie}(78)$

Row 3 = Row 3 * $\text{GCD}(756, 90)$

After getting the new key now use it to decrypt the above cipher text and predict the plain text.

B) (5)

? ? ?
 You know a Cipher that uses 3 * 3 key matrix $K = \begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$ and you
 also know that plain text "CRYPTOGRAPHY" was encrypted to the
 ciphertext "XZQHKUUVENWQ". Determine K.

- 5) Consider this 48-bit input and 4 S boxes as mentioned as part of the question.

A) 101110100101011101101010110110001101111100111000

	6 bit input	Row value	Column value	S box result	4 bit output
S box S1					
S box S2					
S box S3					
S box S4					

- a) Compress input to 32 bit output, represent it as per table.
 b) Invert the leftmost bit of each of the eight 6 bit input blocks.
 Determine 32 bit output and what fraction of your 32 bit output is different from 32 bit output obtained from question (a).

S ₁																
	x0000x	x0001x	x0010x	x0011x	x0100x	x0101x	x0110x	x0111x	x1000x	x1001x	x1010x	x1011x	x1100x	x1101x	x1110x	x1111x
0yyyy0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0yyyy1	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
1yyyy0	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
1yyyy1	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13
S ₂																
	x0000x	x0001x	x0010x	x0011x	x0100x	x0101x	x0110x	x0111x	x1000x	x1001x	x1010x	x1011x	x1100x	x1101x	x1110x	x1111x
0yyyy0	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
0yyyy1	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
1yyyy0	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
1yyyy1	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9
S ₃																
	x0000x	x0001x	x0010x	x0011x	x0100x	x0101x	x0110x	x0111x	x1000x	x1001x	x1010x	x1011x	x1100x	x1101x	x1110x	x1111x
0yyyy0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
0yyyy1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
1yyyy0	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
1yyyy1	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12
S ₄																
	x0000x	x0001x	x0010x	x0011x	x0100x	x0101x	x0110x	x0111x	x1000x	x1001x	x1010x	x1011x	x1100x	x1101x	x1110x	x1111x
0yyyy0	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
0yyyy1	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
1yyyy0	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
1yyyy1	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

(5)

- B) Consider a block cipher using 8-bit blocks that is based on the basic DES architecture (Feistel network) with two rounds and no initial or final permutation. The scrambling function for round i is $f_i(x, K) = (2i \cdot K)^x \bmod 15$, for $i = 1, 2$, where the key K is a member of Z_{15} .

(5)

If $K = 7$ and the ciphertext is 00111111, Examine the plaintext?

-----End-----